ADVANCED LINEAR ALGEBRA
HOMEWORK 1

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P.1.2 Is $[1 \quad 0 \quad 1]^\top \in \mathbb{R}^3$ in the span of $[-1 \quad 3 \quad 2]^\top$ and $[1 \quad 1 \quad -2]^\top$? Why?

Solution. We will show that $[1 \quad 0 \quad 1]^\top \not\subseteq \operatorname{span}([-1 \quad 3 \quad 2]^\top, [1 \quad 1 \quad -2]^\top)$ by contradiction. So assume that there is some $a, b \in \mathbb{R}$ with

$$a \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Which if we expand we get the following equations:

$$\begin{aligned} -a + b &= 1 \\ 3a + b &= 0 \\ 2a - 2b &= 1 \end{aligned}$$

If we simply try to solve the first and third equation we need that the matrix equation to hold:

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

For this to be the case we would need that $\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$ be invertible so its determinant would need to be non-zero. Let's calculate the determinant:

$$\det \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} = (2 - 2) = 0.$$

So our matrix is not invertible, and we cannot find a unique a and b to solve our system of equations. $[1 \quad 0 \quad 1]^\top \not\subseteq \operatorname{span}([-1 \quad 3 \quad 2]^\top, [1 \quad 1 \quad -2]^\top)$ $\square$

P.1.3 Is $[7 \quad 8 \quad 9]^\top \in \mathbb{R}^3$ in the span of $[1 \quad 2 \quad 3]^\top$ and $[4 \quad 5 \quad 6]^\top$? Why? Is $[10 \quad 11 \quad 12]^\top$?

Solution. In a similar fashion to problem **P.1.2**, we choose an $a, b \in \mathbb{R}$, and we set:

$$a \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + b \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

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This reduces to the equations:

$$\begin{aligned} a + 4b &= 7 \\ 2a + 5b &= 8 \\ 3a + 6b &= 9 \end{aligned}$$

So if we let $a = -4b + 7$, as given by the first equation and then substitute into the second equation, we see that

$$\begin{aligned} 2(-4b + 7) + 5b &= 8 \\ -8b + 14 + 5b &= 8 \\ -3b &= -6 \\ b &= 2 \end{aligned}$$

Substituting back into our top equality we see that $a + 4(2) = 7$, thus $a = -1$. Furthermore if we plug these values back into our original linear combination, we get that

$$\begin{aligned} -1\begin{bmatrix}1\\2\\3\end{bmatrix} + 2\begin{bmatrix}4\\5\\6\end{bmatrix} &= \begin{bmatrix}7\\8\\9\end{bmatrix} \\ \begin{bmatrix}-1\\-2\\-3\end{bmatrix} + \begin{bmatrix}8\\10\\12\end{bmatrix} &= \begin{bmatrix}7\\8\\9\end{bmatrix} \end{aligned}$$

If we were to do the same inspection for the vector $[10 \quad 11 \quad 12]^\top$ we write $a, b \in \mathbb{R}$ with

$$a\begin{bmatrix}1\\2\\3\end{bmatrix} + b\begin{bmatrix}4\\5\\6\end{bmatrix} = \begin{bmatrix}10\\11\\12\end{bmatrix},$$

We then expand to:

$$\begin{aligned} a + 4b &= 10 \\ 2a + 5b &= 11 \\ 3a + 6b &= 12 \end{aligned}$$

So we have that $a = -4b + 10$ which if we plug into the second equation we get:

$$\begin{aligned} 2(-4b+10)+5b &= 11 \\ -8b+20+5b &= 11 \\ -3b &= -9 \\ b &= 6 \end{aligned}$$

And if we plug b back in we get $a = -24 + 10 = -14$. So we test this out to see if it works:

$$\begin{aligned} -14\begin{bmatrix}1\\2\\3\end{bmatrix} + 6\begin{bmatrix}4\\5\\6\end{bmatrix} &= \begin{bmatrix}10\\11\\12\end{bmatrix} \\ \begin{bmatrix}-14\\-28\\-42\end{bmatrix} + \begin{bmatrix}24\\30\\36\end{bmatrix} &= \begin{bmatrix}10\\2\\-6\end{bmatrix} \neq \begin{bmatrix}10\\11\\12\end{bmatrix} \end{aligned}$$

So $[10 \quad 11 \quad 12]^\top \not\subseteq \mathrm{span}([1 \quad 2 \quad 3]^\top, [4 \quad 5 \quad 6]^\top)$ $\square$

P.1.8 Consider the followoing lists of vectors in $\mathbb{R}^3$:

$$\beta = \begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\s\\6\end{bmatrix}, \begin{bmatrix}7\\8\\9\end{bmatrix},$$

and

$$\gamma = \begin{bmatrix}4\\2\\6\end{bmatrix}, \begin{bmatrix}1\\t\\-1\end{bmatrix}, \begin{bmatrix}2\\-2\\3\end{bmatrix},$$

For what values of s is β linearly dependent. For what values of t is γ linearly dependent.

Solution. (β) We know that β is linearly dependent when $\det(M_\beta) = 0$, where M_β represents the matrix with the vectors of β as its columns. So let us compute the following:

$$\begin{aligned}
\det \begin{bmatrix} 1 & 4 & 7 \\ 2 & s & 8 \\ 3 & 6 & 9 \end{bmatrix} &= 0 \\
1 \cdot \det \begin{bmatrix} s & 8 \\ 6 & 9 \end{bmatrix} + 4 \cdot \det \begin{bmatrix} 2 & 8 \\ 3 & 9 \end{bmatrix} + 7 \cdot \det \begin{bmatrix} 2 & s \\ 3 & 6 \end{bmatrix} &= 0 \\
1(9s - 48) + 4(18 - 24) + 7(12 - 3s) &= 0 \\
9s - 48 + 72 - 96 + 84 - 21s &= 0 \\
-12s + 12 &= 0 \\
s &= 1
\end{aligned}$$

So when $s = 1$ β is linearly dependent.

(γ) Let M_γ be the matrix comprised of the vectors of γ as columns. Similarly to above, we attempt to compute $\det(M_\gamma) = 0$. So we expand:

$$\begin{aligned}
\det \begin{bmatrix} 4 & 1 & 2 \\ 2 & t & -2 \\ 6 & -1 & 3 \end{bmatrix} &= 0 \\
4 \cdot \det \begin{bmatrix} t & -2 \\ -1 & 3 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 2 & -2 \\ 6 & 3 \end{bmatrix} + 2 \cdot \det \begin{bmatrix} 2 & t \\ 6 & -1 \end{bmatrix} &= 0 \\
4(3t - 2) + (6 + 12) + 2(-2 - 6t) &= 0 \\
12t - 8 + 6 + 12 - 4 - 12t &= 0 \\
6 &\neq 0
\end{aligned}$$

Because our terms $12t$ and $-12t$ cancel out and we get that $\det(M_\gamma) = 6$, we have that γ is linearly independent for any value of t. □

P.1.12 Show that the set of $n \times n$ Hermitian matricies is a real vector space.

Proof. Let $\mathcal{H}_n \subseteq \mathbf{M}_n(\mathbb{C})$ be the Hermitian. So by definition for $A \in \mathcal{H}_n$, we have that $A^* = A$. Now we need to prove the vector space axioms using $\mathbb{R}$ as our set of scalars. Begin by setting $A, B \in \mathcal{H}_n$ and $c \in \mathbb{R}$.

Closure under addition. We simply perform some easy algebraic manipulation beginning with the conjugate transpose of the addition $A + B$

$$(A + B)^* = A^* + B^* = A + B \in \mathcal{H}_n.$$

Scalar multiplication. For this vector space axiom, we begin with $(cA)^*$ and perform algebraic manipulation as follows:

$$(cA)^* = \bar{c}A^* = cA^* = cA \in \mathcal{H}_n$$

since $\bar{c} = c$ by $c \in \mathbb{R}$.

The zero vector. This is trivially true as

$$\hat{0}^* = \hat{0} \in \mathcal{H}_n$$

Note. Notice $\mathcal{H}_\backslash$ is not a complex vector space because $(iA)^* = -iA^* = -iA \neq iA$. $\square$

P.1.13 Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(a) Choose two of the following and show that each is a factorization $A = XY$, in which the columns of X and rows of Y are each linearly independent:

$$\begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 5 & 6 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 3 \\ 4 & 6 \\ 7 & 9 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix}, \begin{bmatrix} -1 & 3 \\ -1 & 9 \\ -1 & 15 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

(b) Which of the decompositions is the one described by Theorem 1.7.5? Why?

Solution. (a) Let's simply expand the first two decompositions XY to see if $XY = A$:

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix} &= \begin{bmatrix} 1+0(2) & 1(0)+2 & -1+4 \\ 4+0(5) & 0(4)+1(5) & -4+10 \\ 7+0(8) & 0(7)+1(8) & -7+16 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \end{aligned}$$

and

$$\begin{aligned} \begin{bmatrix} 2 & 3 \\ 5 & 6 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} &= \begin{bmatrix} 4-3 & 2+3(0) & 0(2)+3(1) \\ 2(5)-1(6) & 1(5)+0(6) & (0)5+1(6) \\ 2(8)-1(9) & 1(8)+0(9) & 0(8)+1(9) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \end{aligned}$$

(b) We have that expansions 1,2, and 3 above are the pivot column decompositions because for each X in the decomposition $A = XY$, if we write $X = [\hat{x}_1 \quad \hat{x}_2]$ we have that for each of the X's, $\hat{x}_1, \hat{x}_2 \in \text{col}(A)$. □

P.1.27 (a) Show that $\text{col}(A) = \mathbb{R}^3$ for

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a_3} \quad \mathbf{a}_4] = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \in \mathbf{M}_n(\mathbb{R})$$

(b) Find $i, j, k \in \{1, 2, 3, 4\}$ such that $\text{span}\{\mathbf{a}_i, \mathbf{a}_j, \mathbf{a}_k\} = \mathbb{R}^3$. And find distinct $i, j, k \in \{1, 2, 3, 4\}$ such that $\text{span}\{\mathbf{a}_i, \mathbf{a}_j, \mathbf{a}_k\}$.

Solution. (a) *Claim.* $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a_3}\}$ are linearly independent thus $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\} = \mathbb{R}^3$. So we set $X = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a_3}]$, and we want to show that $\det(X) \neq 0$. So we expand in the following way:

$$\begin{aligned} \det[X] &= \det \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \\ &= 1 \cdot \det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ &= 1 + 0 + 0 \\ &= 1 \neq 0. \end{aligned}$$

So we see that $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a_3}\}$ are linearly independent which implies that $\text{col}(X) = \text{col}(A) = \mathbb{R}^3$

(b) For linear dependence, consider the column space $X = [\mathbf{a}_1 \quad \mathbf{a}_3 \quad \mathbf{a_4}]$. We want to see that $\det[X] = 0$. So we compute

$$\begin{aligned} \det[X] &= \det \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \\ &= 1 \cdot \det \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \\ &= 0 + 0 + 0 \\ &= 0. \end{aligned}$$

So $[\mathbf{a}_1 \quad \mathbf{a}_3 \quad \mathbf{a_4}]$ are not linearly independent, and $\text{span}[\mathbf{a}_1 \quad \mathbf{a}_3 \quad \mathbf{a_4}] \neq \mathbb{R}^3$. And, in fact we could show $\text{span}[\mathbf{a}_1 \quad \mathbf{a}_3 \quad \mathbf{a_4}] = \mathbb{R}^2$. □